

A) Building basic classes

Exercise A1) Person:

Create a module called **people.py**. Within this module, create a **Person** class.

A Person should contain 4 attributes (variables):

1. First name
2. Second name
3. Age
4. Whether or not they're left-handed (True/False)

You can set the default values of these to be anything you want.

Commit your code after your class is complete.

Exercise A2) Accessing an object's attributes:

In a separate module (**people_testbed.py**), write a program (main script) that:

1. Imports your Person class from the people.py module.
2. Creates a Person object (you will need to use `variable_name = Person()` to do this)
3. Displays the Person's full name in a formatted manner based on their left-handedness status
 - a. If they are left-handed, the text should be displayed normally.
 - b. If they are not left-handed, the text should be displayed in all caps.

Exercise A3) Setting/changing an object's attributes:

Amend your program from A2 to create a second Person object (p2). Set the data inside to user-supplied values (i.e. ask the user to enter each of the values and store them within p2's attributes/variables)

Once the user has supplied all the values required, display their full name in the same formatted manner as used for your Person from A2, i.e. display their name in all caps if they are left-handed, and display it normally otherwise.

Commit your code after your program is complete.

B) Adding methods

Reminder: Whenever you create a method in a class, you need to:

1. Take in *self* as the **first** parameter
2. Refer to class attributes as **self.attribute_name**, not just the attribute name directly.

Exercise B1) Rectangle:

Create a module called **shapes.py**. Within this module, create a **Rectangle** class.

A Rectangle should contain 3 attributes (variables):

1. Length
2. Width
3. Colour

You can set the default values of these to be anything you want.

Test your Rectangle class by creating a rectangle instance in a main script (at the bottom of the shapes.py file) and displaying its data to the console.

Exercise B2) Creating a display method:

Create a method (**display()**) within your Rectangle class to display its information. This method should print the values of the attributes in the format:

"Rectangle[length=X, width=Y, colour=Z]" where X, Y and Z are the values of the three attributes.

Reminder: We use the same syntax to create a method as we do to create a function. They're different names for essentially the same thing.

Test your display() method by calling it on the rectangle in your main script from part B1.

Commit your code after your class is complete, including your method.

Exercise B3) Calculating the area of a rectangle:

Create a method (***calc_area()***) within your Rectangle class to calculate its area (length * width) and return it.

Test your ***calc_area()*** method by calling it in your main script from part B1 and displaying the result.

Commit your code after your method is complete and functions correctly.

Exercise B4) Using your class:

Remember to import your Rectangle class!

Write a program that asks a user to create 5 Rectangle objects & stores them in a list. For each, your program should:

1. Create the current Rectangle
2. Ask the user for this Rectangle's length, width and colour.
3. Set these values in the current rectangle.
4. Add this Rectangle to the list.

Once your list contains 5 Rectangles, your program should:

- Locate the Rectangle with the largest area and display its details using its display method.
- Locate the Rectangle with the smallest width and display its position in the list.
- Locate all Rectangles with the colour "red" (case-insensitive), store them in a new list and display them once all Rectangles have been processed. If there are no red Rectangles, display a message indicating this to the user.
- Ask the user to enter a colour, then repeat the above action to find, store and display all Rectangles of that colour.

Note: You should consider how you might use functions to improve your code.

Commit your code:

- When the user can create rectangles
- When your program can search for rectangles based on different characteristics